

**K S Tharunpranav 251801359-2025-AIDS** ▾**K2****Started on** Monday, 24 August 2026, 9:00 PM**State** Finished**Completed on** Monday, 24 August 2026, 9:00 PM**Time taken** 28 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Given two arrays `array_One[]` and `array_Two[]` of same size `N`. We need to first rearrange the arrays such that the sum of the product of pairs (1 element from each) is minimum. That is $\text{SUM}(A[i] * B[i])$ for all `i` is minimum.

For example:

Input	Result
3	28
1	
2	
3	
4	
5	
6	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int compare_asc(const void *a, const void *b) {
5      long long arg1 = *(const long long *)a;
6      long long arg2 = *(const long long *)b;
7      if (arg1 < arg2) return -1;
8      if (arg1 > arg2) return 1;
9      return 0;
10 }
11
12 int compare_desc(const void *a, const void *b) {
13     long long arg1 = *(const long long *)a;
14     long long arg2 = *(const long long *)b;
15     if (arg1 < arg2) return 1;
16     if (arg1 > arg2) return -1;
17     return 0;
18 }
19
20 int main() {
21     int n;
22     if (scanf("%d", &n) != 1) return 0;
23
24     long long *array_One = (long long *)malloc(n * sizeof(long long));
25     long long *array_Two = (long long *)malloc(n * sizeof(long long));
26     if (array_One == NULL || array_Two == NULL) return 0;
27
28     for (int i = 0; i < n; i++) {
29         if (scanf("%lld", &array_One[i]) != 1) return 0;
30     }
31
32     for (int i = 0; i < n; i++) {
33         if (scanf("%lld", &array_Two[i]) != 1) return 0;
34     }
35
36     qsort(array_One, n, sizeof(long long), compare_asc);
37     qsort(array_Two, n, sizeof(long long), compare_desc);
38
39     long long min_sum = 0;
40     for (int i = 0; i < n; i++) {
41         min_sum += array_One[i] * array_Two[i];
42     }
43
44     printf("%lld\n", min_sum);
45
46     free(array_One);
47     free(array_Two);
48     return 0;
49 }
50

```

	Input	Expected	Got	
✓	3 1 2 3 4 5 6	28	28	✓
✓	4 7 5 1 2 1 3 4 1	22	22	✓
✓	5 20 10 30 10 40 8 9 4 3 10	590	590	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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